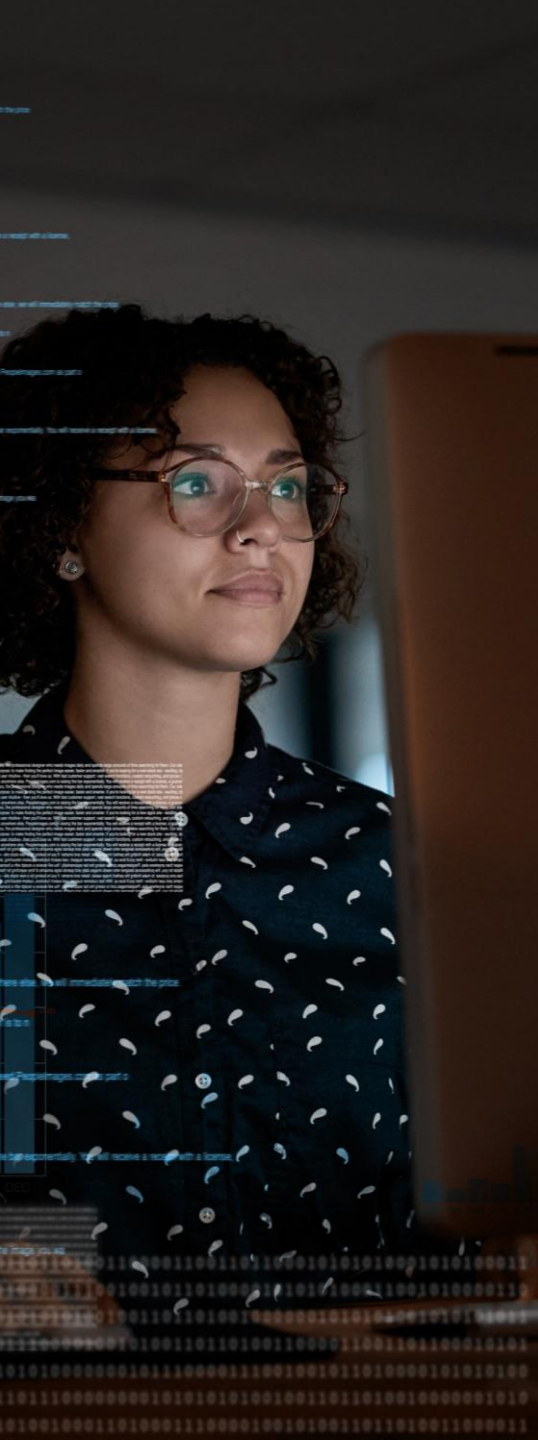




PMI-ACP® EXAM PREP

PMI Authorized Training Partner

BOOTCAMP

Session 8

- This course will assist learners in preparing for PMI's PMI-ACP Exam (2024 Update)

ATTENDANCE TRACKING

Percipio Users:

Name is based on your log in information in Percipio

Using Zoom:

Enter your first and last name

BREAKS



Yes! We will have periodic breaks

For attendance purposes, please stay logged in during all breaks.



CHAT vs Q&A

Please use the **Chat** for:

- **Greetings** before the session starts and during breaks
- Once the session starts , the chat may be closed or changed to *Hosts & Panelists Only* to minimize disruptions and focus on important information.
- The instructor may open the chat during the session for student **to respond to the instructor's questions** and create a group dialog.

CHAT vs Q&A

Please use the **Q&A** for:

- **Technical assistance** – Begin with: Percipio or Non-Percipio student
- Guidance on how to **access course material** – Begin with: Percipio or Non-Percipio
- Clarification and **questions on lecture points**, if not answered by instructor
- The Q&A may be open and closed throughout the session to allow us to address questions/issues in a timely manner.
- **Please be very patient, the support team responds to many inquiries per session**

IS LIVE ATTENDANCE REQUIRED?

- **YES**, if you are taking this training to register for the PMI-ACP exam
- You are **allowed to miss up to two sessions** IF you make up the sessions by **watching the video replays**.
- A **missed** session means you were disconnected for **more than a total of 15 mins** for the duration of the session.
- *Please see the Bootcamp Calendar for information about upcoming sessions at: <http://calendar.skillsoft.com/>



ACCESSING THE VIDEO REPLAYS

1. Go to: <https://github.com/Skillsoft-Content/PMI-ACP-Replay>
2. Replays will be available within 2 business days after the session ends.
3. Click on the Excel file for the year you attended the Bootcamp. You won't see a *file open* option, but it is selected.
4. Click the *Download raw file* button on the far left-hand side.
5. Open the downloaded file using this password: [acpB00tcampReplay!](#)



Those are zero's not the letter O. The password is case sensitive.

6. Locate and open the worksheet tab that corresponds with the bootcamp you attended
7. Make a note of the passcode.
8. Paste the provided link into your browser.
9. Complete the required registration steps
10. Input the passcode when prompted
The password to open the Excel file is NOT the passcode to access the replay.

Note: Replays will be available for 1 year.
They are not available for download.

No limit to watch replays to study

PMI®-Authorized PMI Agile
Certified Practitioner (PMI-ACP)®
Exam Preparation Course

Lesson Four

Delivery

Version 1.0 | 2024 Release



During this session



**Seek early
feedback**



**Manage
agile metrics**



**Manage
impediments
and risk**



**Recognize
and eliminate
waste**



**Perform
continuous
improvements**



**Engage
customers**



**Optimize
flow**



**Obtain metrics
and feedback
to drive
continuous
improvements**



**Implement
improvement
actions**



**Evaluate the
effectiveness
of process
improvement**



**Seek early
feedback**



**Manage
agile metrics**



**Manage
impediments
and risk**



**Recognize
and eliminate
waste**



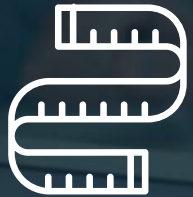
**Perform
continuous
improvements**



**Engage
customers**



**Optimize
flow**



Obtain metrics and feedback to drive continuous improvements

Section 1 of 3

Are our process efficiency metrics improving?

Cycle time

Lead time



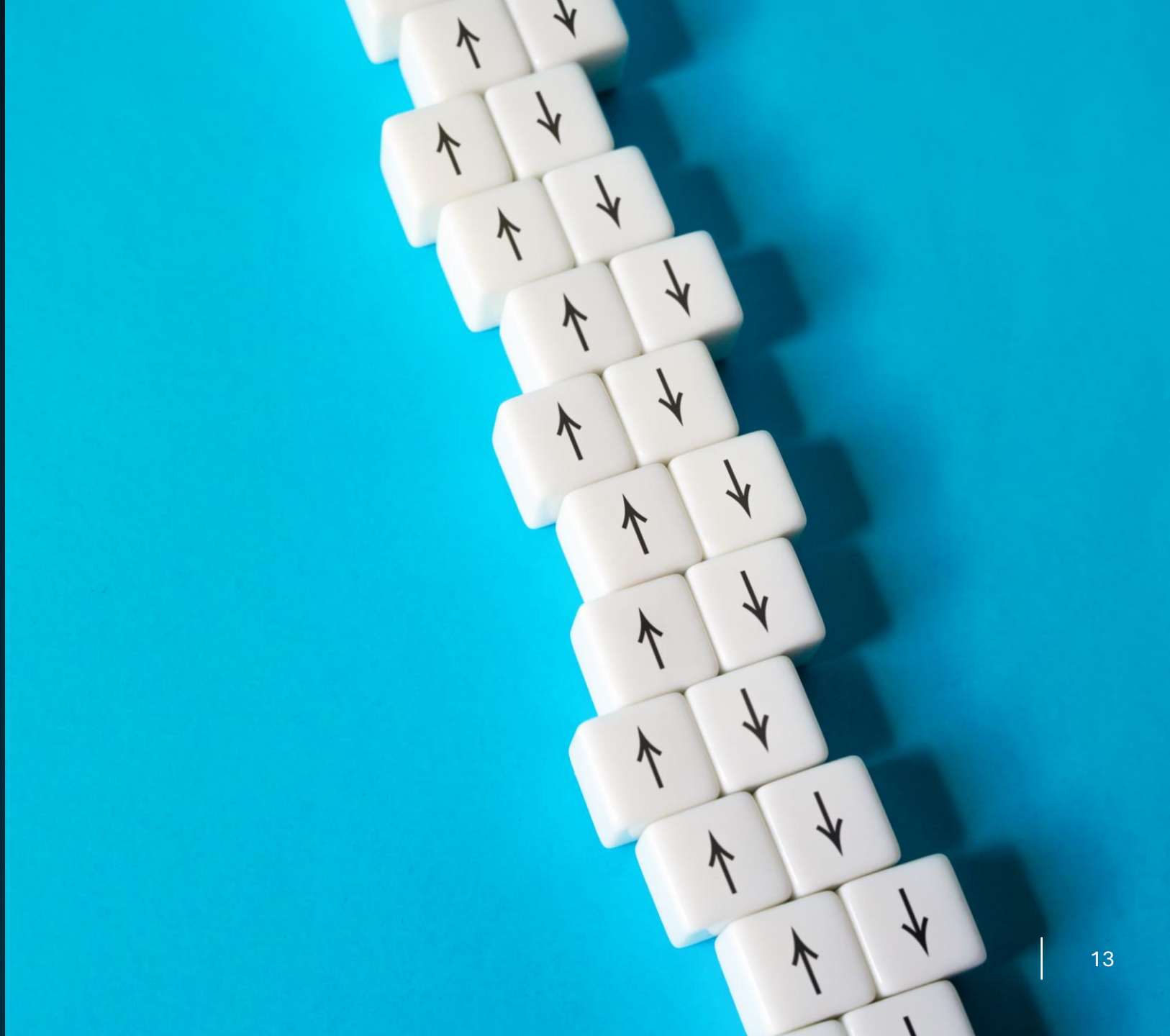
Improving lead time and cycle time

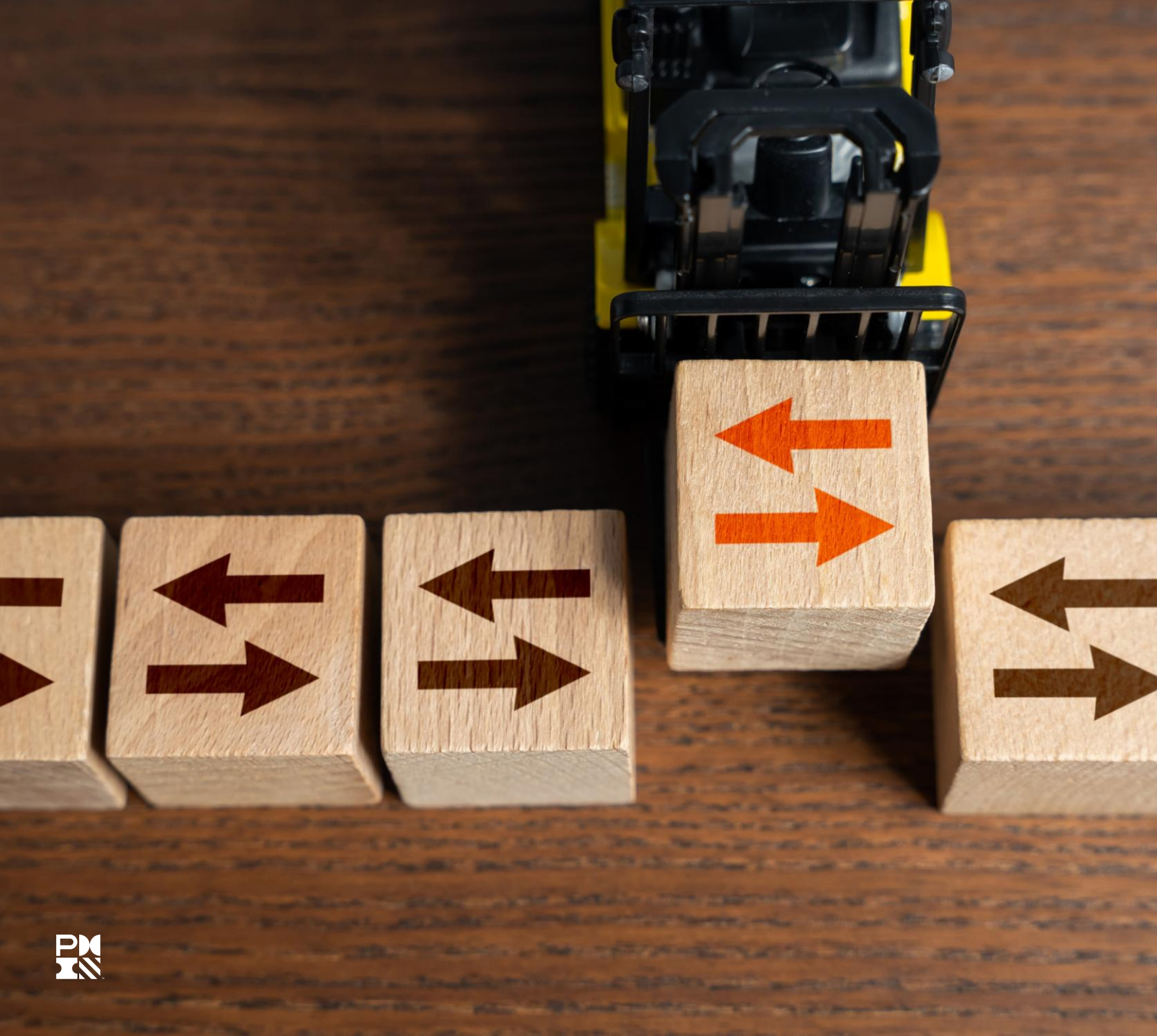
We want lead and cycle
times to decrease

Are our throughput metrics improving?

Features complete versus
features remaining

Velocity





Improving features complete versus features remaining and velocity

We want the ratio of features complete to features remaining to increase

We want velocity to increase

Beyond efficiency and throughput



**Needs of our
customers**



**Needs of our team
members**

What metrics should we use to measure these?

Customer satisfaction metrics



Customer satisfaction score

Net promoter score[®]

Team satisfaction metric

Employee net promoter score[®]



**“How likely is it that you would recommend
our team as a place to work?”**

Knowledge check

Which of these metrics would be best for tracking process efficiency?

A. Features complete vs. features remaining

B. Customer satisfaction score

C. Net promoter score

D. Cycle time

Of these metrics, **cycle time** would be the best for tracking process efficiency.

Knowledge check

Which of these metrics would be best for tracking throughput?

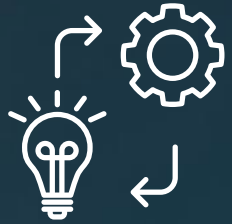
A. Features complete versus features remaining

B. Customer satisfaction score

C. Net promoter score

D. Cycle time

Of these metrics **features complete versus features remaining** would be the best for tracking throughput.



Implement improvement actions

Section 2 of 3



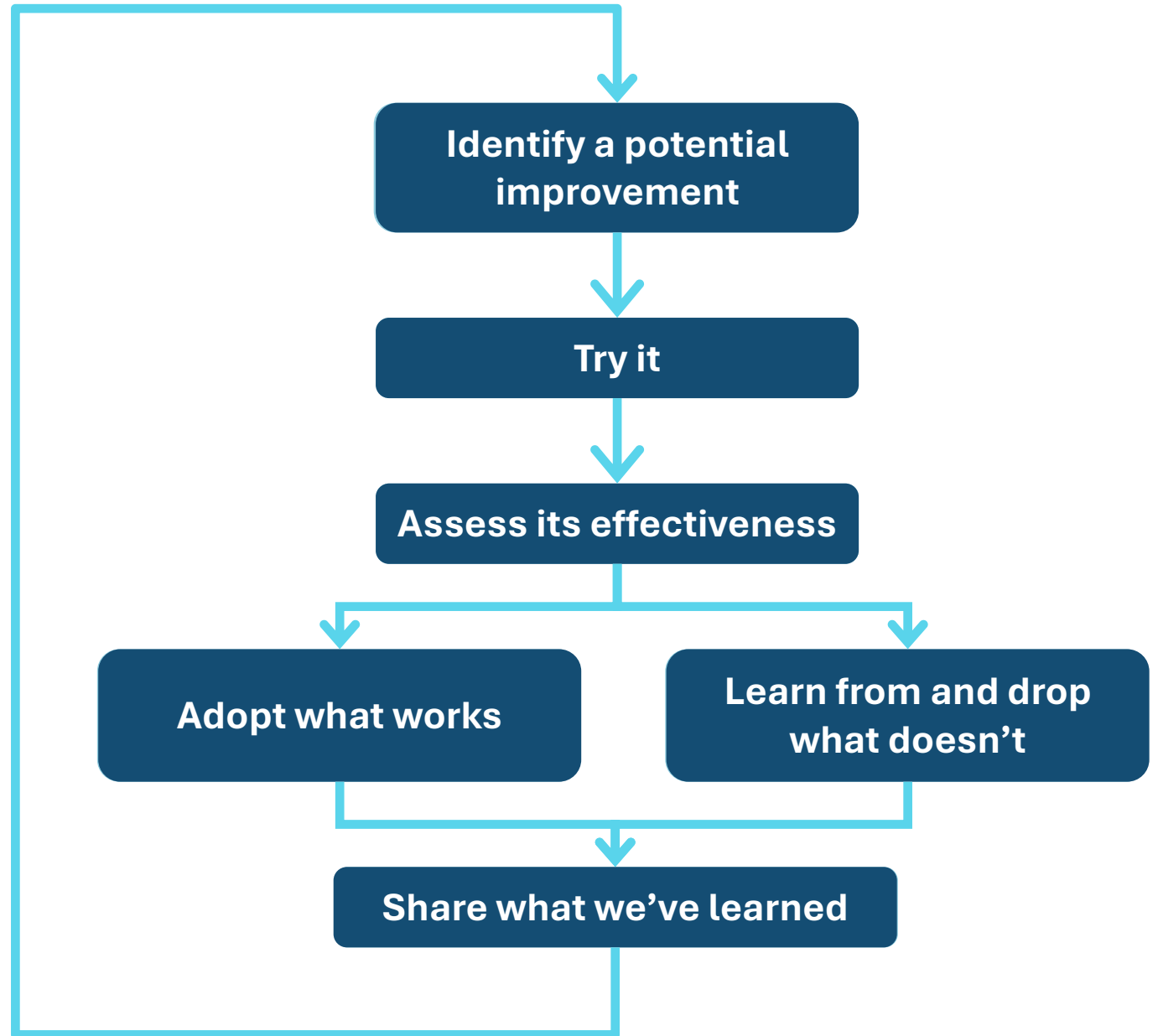


Run experiments

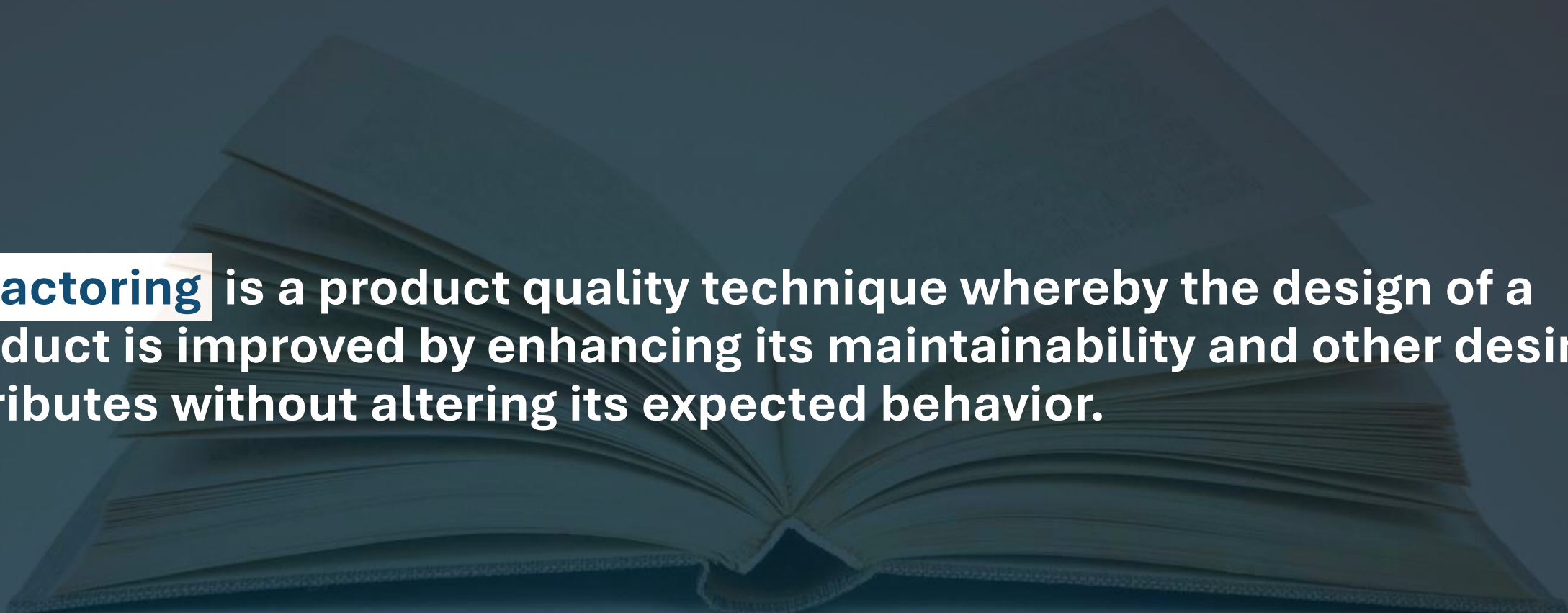
Don't do it randomly

You need a
systematic approach

A systematic approach for improving through experiments



Refactoring



Refactoring is a product quality technique whereby the design of a product is improved by enhancing its maintainability and other desired attributes without altering its expected behavior.

Add refactoring tasks to the backlog

Explain the impacts of the refactoring work to the product owner so they can prioritize it appropriately

Plan for continuing refactoring



Kanban

A means to design, manage, and improve flow for knowledge work.

It allows teams to start where they are to drive evolutionary change.





Improving our process

If we can be agile in the way we implement agile, the full range of agile practices becomes a powerful toolbox

That's the idea behind Disciplined Agile® (DA®)

DA Browser (<https://dabrowser.pmi.org/>)



DA Browser

DA 5.5

Disciplined Agile Delivery (DAD)

Inception

- Form Team
- Align With Enterprise Direction
- Explore Scope
- Identify Architecture Strategy
- Plan the Release
- Develop Test Strategy
- Develop Common Vision
- Secure Funding

Construction

- Prove Architecture Early
- Address Changing Stakeholder Needs**
- Produce a Potentially Consumable Solution
- Improve Quality
- Accelerate Value Delivery

Transition

- Ensure Production Readiness
- Deploy the Solution

Ongoing

- Grow Team Members
- Coordinate Activities
- Address Risk
- Evolve Way of Working (WoW)
- Leverage and Enhance Existing Infrastructure

Address Changing Stakeholder Needs

v5.2

Stakeholder Interaction With Team

Elicit Requirements

Explore Stakeholder Needs

Active stakeholder participation
On-site customer
Indirectly via product owner
Indirectly via business analyst
Indirectly via product manager
Indirectly via digital means
Change control board (CCB)

All-hands demos
Interviews
Iteration demos
Just-in-time (JIT) model storming
Look-ahead modeling/backlog refinement
On-demand demos

Active stakeholder participation/on-site customer
Agile Modeling session/big room planning
Behavior-driven development (BDD)
Definition of ready (DoR)
Design sprint (User experience (UX))
Detailed requirements specification
High-level requirements specification
Just-in-time (JIT) model storming
Look-ahead modeling/backlog refinement
Split (A/B) testing

 COPY URL

Address Changing Stakeholder Needs

ADDITIONAL RESOURCES

The Address Changing Stakeholder Needs process goal provides options for DA teams to react to changing needs effectively. Change happens. Sometimes a change is a completely new piece of work, sometimes it's a modification to work you haven't started yet, sometimes it's a modification to work you're currently doing, and sometimes it's a modification to work you've already delivered.

Of course, new information isn't always a requirement change. The reality is that as a team works on something the stakeholder's understanding, and in turn the team's understanding, of the true requirements will evolve and new or changed details will surface. In an effort to maintain a sustainable pace we have seen some 'purist' team leads disallow new requirement details to be brought into an iteration to motivate product owners to do a better job of look-ahead modeling. In these situations they ask the product owner to create a new work item and add it to the backlog to be estimated and prioritized for development in a future iteration. Obviously this doesn't help to build a good working relationship between the business and the delivery team. A better approach is for the team to expect details to emerge during the iteration, often via just in time (JIT) model storming or impromptu feedback sessions/demos, and ensure that they allocate a buffer as a contingency during their iteration planning session. When new information about an existing work item proves to be too large, at that point the team can ask the product owner to introduce new work items. These decisions are described as options in the Accept Changes decision point.

Why This is Important

DA Browser (<https://dabrowser.pmi.org/>)

The screenshot displays the DA Browser interface. On the left, a sidebar lists the 'Disciplined Agile Delivery (DAD)' framework, categorized into Inception, Construction, Transition, and Ongoing. The 'Address Changing Stakeholder Needs' process goal is highlighted in the sidebar. The main content area shows a diagram of this process goal, which branches into three sub-goals: 'Stakeholder Interaction With Team', 'Elicit Requirements', and 'Explore Stakeholder Needs'. To the right of the diagram, a list of activities is provided for each sub-goal. An orange arrow points from the text 'Process goals' to the 'Address Changing Stakeholder Needs' process goal in the diagram.

DA 5.5

Disciplined Agile Delivery (DAD)

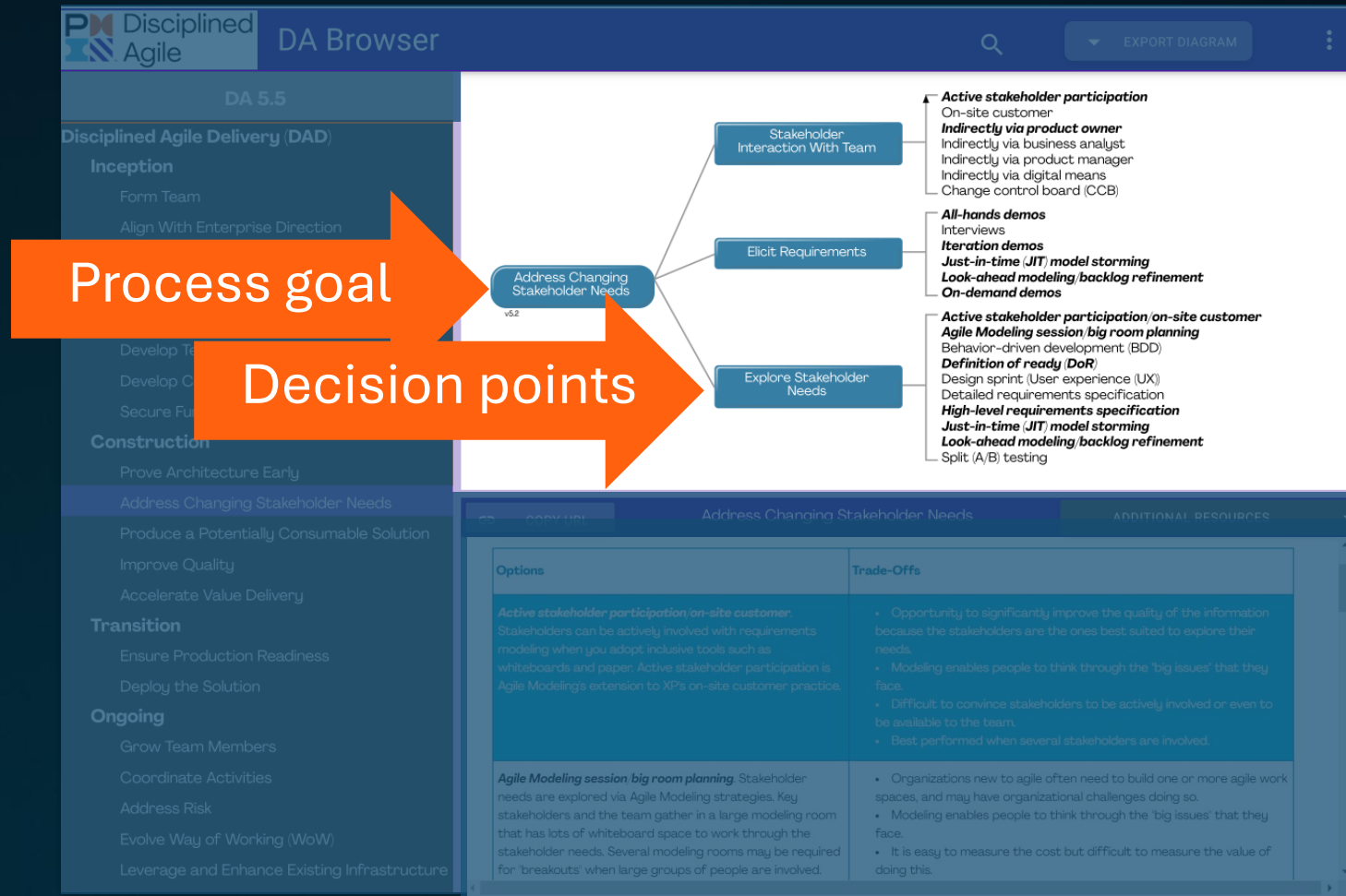
- Inception**
 - Form Team
 - Align With Enterprise Direction
 - Explore Scope
 - Identify Architecture Strategy
 - Plan the Release
 - Develop Test Strategy
 - Develop Common Vision
 - Secure Funding
- Construction**
 - Prove Architecture Early
 - Address Changing Stakeholder Needs**
 - Produce a Potentially Consumable Solution
 - Improve Quality
 - Accelerate Value Delivery
- Transition**
 - Ensure Production Readiness
 - Deploy the Solution
- Ongoing**
 - Grow Team Members
 - Coordinate Activities
 - Address Risk
 - Evolve Way of Working (WoW)
 - Leverage and Enhance Existing Infrastructure

Address Changing Stakeholder Needs

- Stakeholder Interaction With Team**
 - Active stakeholder participation
 - On-site customer
 - Indirectly via product owner
 - Indirectly via business analyst
 - Indirectly via product manager
 - Indirectly via digital means
 - Change control board (CCB)
- Elicit Requirements**
 - All-hands demos
 - Interviews
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 - Just-in-time (JIT) model storming
 - Look-ahead modeling/backlog refinement
 - On-demand demos
- Explore Stakeholder Needs**
 - Active stakeholder participation/on-site customer
 - Agile Modeling session/big room planning
 - Behavior-driven development (BDD)
 - Definition of ready (DoR)
 - Design sprint (User experience (UX))
 - Detailed requirements specification
 - Just-in-time (JIT) model storming
 - Look-ahead modeling/backlog refinement
 - Testing

Why This is Important

DA Browser (<https://dabrowser.pmi.org/>)



DA Browser (<https://dabrowser.pmi.org/>)

The screenshot displays the DA Browser interface. On the left, a sidebar lists the Disciplined Agile Delivery (DAD) framework components: Inception, Construction, and Transition. The main area shows a diagram titled 'Address Changing Stakeholder Needs' (v5.2). This diagram illustrates the process of eliciting requirements through stakeholder interaction with the team. A large orange arrow labeled 'Options' points from the diagram to a table below. The table is titled 'Options and trade-offs' and contains two columns: 'Options' and 'Trade-Offs'.

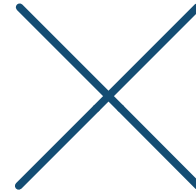
Options	Trade-Offs
Active stakeholder participation/on-site customer. Stakeholders can be actively involved with requirements modeling when you adopt inclusive tools such as whiteboards and paper. Active stakeholder participation is Agile Modeling's extension to XP's on-site customer practice.	• Opportunity to significantly improve the quality of the information because the stakeholders are the ones best suited to explore their needs. • Modeling enables people to think through the 'big issues' that they face. • Difficult to convince stakeholders to be actively involved or even to be available to the team. • Best performed when several stakeholders are involved.
Agile Modeling session/big room planning. Stakeholder needs are explored via Agile Modeling strategies. Key stakeholders and the team gather in a large modeling room that has lots of whiteboard space to work through the stakeholder needs. Several modeling rooms may be required for 'breakouts' when large groups of people are involved.	• Organizations new to agile often need to build one or more agile work spaces, and may have organizational challenges doing so. • Modeling enables people to think through the 'big issues' that they face. • It is easy to measure the cost but difficult to measure the value of doing this.

Knowledge check

True or false: Refactoring only applies to software projects



True



False

Knowledge check

True or false: If you want consistent and sustained improvement you need a systematic approach



True



False



Evaluate the effectiveness of process improvement

Section 3 of 3

Running experiments

Experiments
in the way you work

Experiments
in the work itself





Following up on these experiments at the next retrospective

If it worked, share it

If it didn't work, learn from it

How do we know if it worked?

We measure the results



Measuring process efficiency using a value stream map



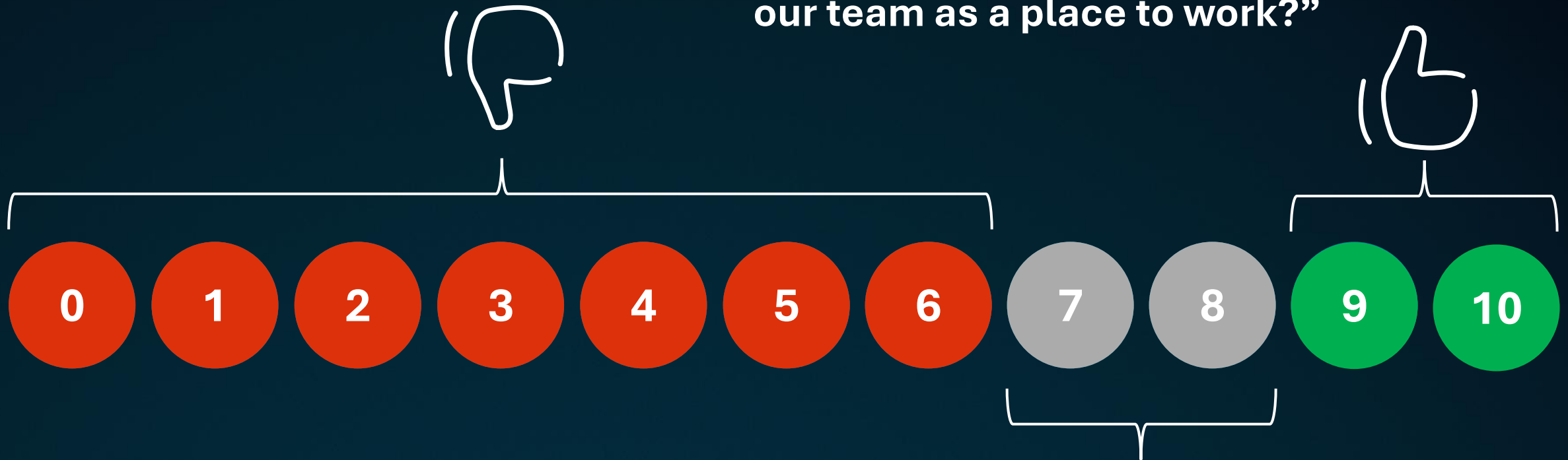
Formula for process efficiency



$$\text{Process efficiency} = \text{Total value-added time} \div \text{Total cycle time}$$

Measuring team engagement

“How likely is it that you would recommend
our team as a place to work?”



Employee net promoter score

Measuring team engagement



Calculating your net promoter score[®]

$$\text{😊} \% - \text{😡} \% = \text{NPS}$$



Asking for external reviews and audits

Clear objectives and scope

Confidentiality

Team buy-in

Actionable feedback

Implementation
and follow-up

Working with your PMO, VDO, or ATO

Clear goals and objectives

Stakeholder engagement

Assessment and analysis

Action plan





Using a community of practice

Relevant experience

Active and engaged

Communication
and collaboration

Ask other teams
about their most
valuable improvements

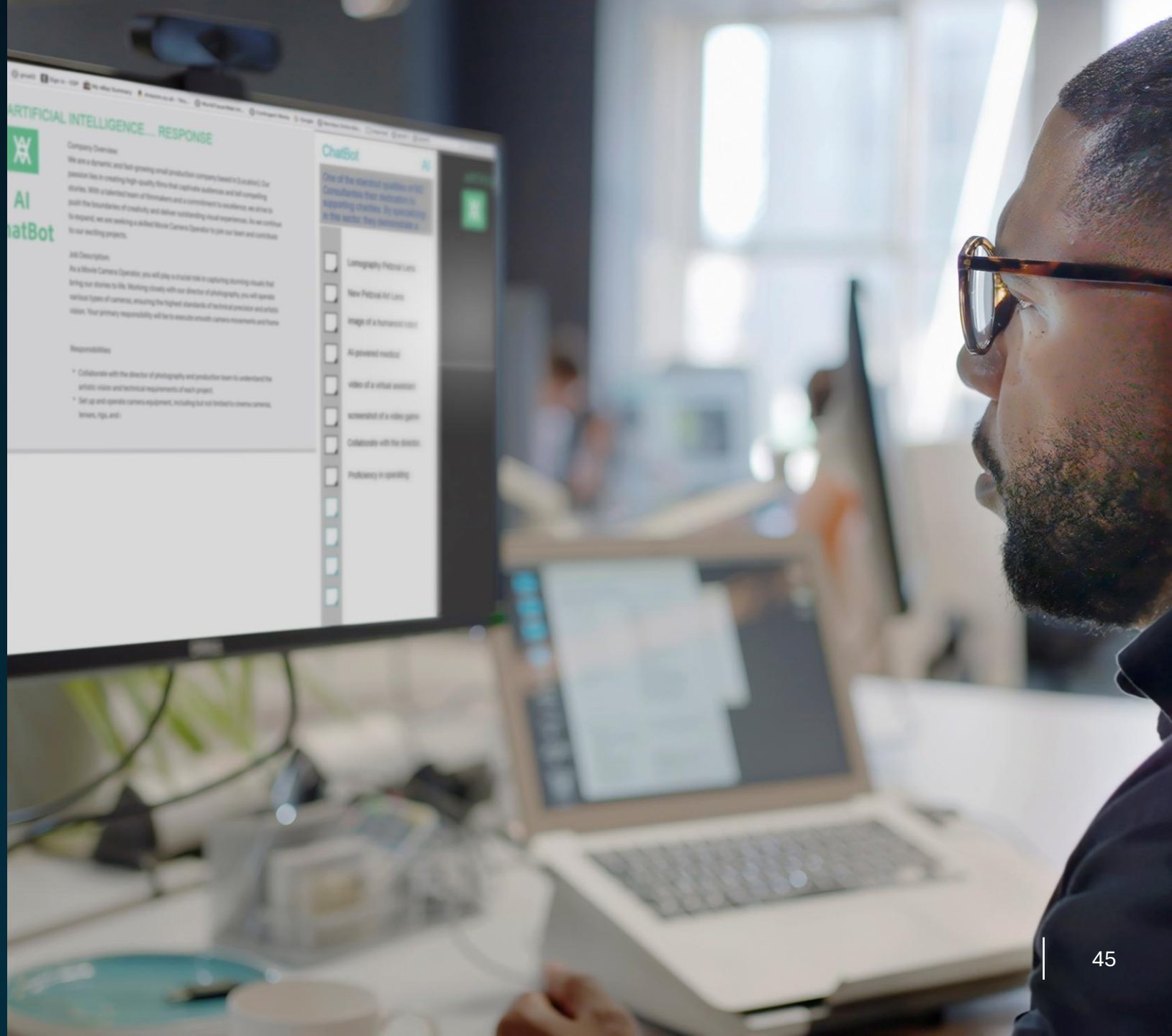
Using artificial intelligence

Targeted improvement suggestions

Understand the limitations

Ensure data privacy and security

Balance automation with human expertise



Knowledge check

When trying out a new way of working, where should the team evaluate whether it is working?

A. During a daily coordination meeting

B. With the project management office (PMO)

C. After measuring team engagement

D. At the next retrospective

When trying out a new way of working, the team should evaluate whether it is working **at the next retrospective.**



**Identify and
analyze
customer
needs**



**Validate that
iteration
deliverables
meet
acceptance
criteria**



**Encourage
collaboration
between
customer and
team**



**Seek early
feedback**



**Manage
agile metrics**



**Manage
impediments
and risk**



**Recognize
and eliminate
waste**



**Perform
continuous
improvements**



**Engage
customers**



**Optimize
flow**



Identify and analyze customers and their needs

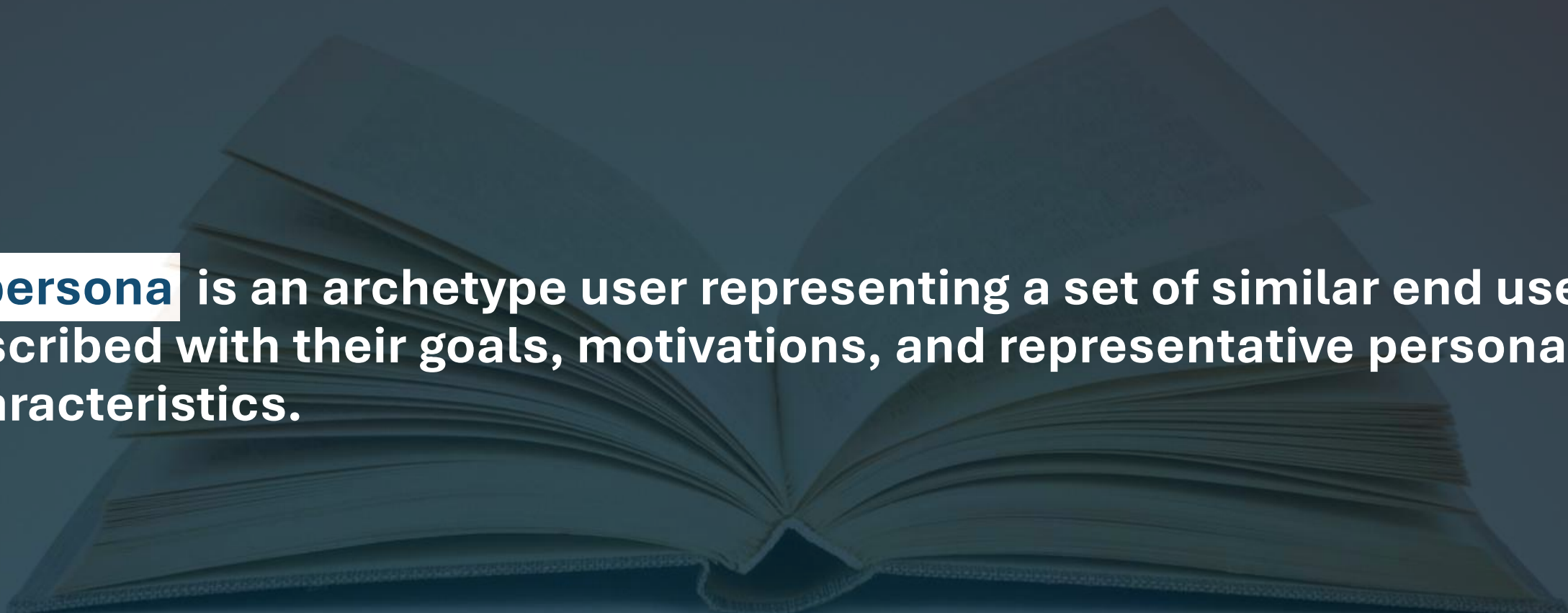
Section 1 of 3

Persona modeling

Doesn't replace
requirements—
it enhances them



Persona



A **persona** is an archetype user representing a set of similar end users described with their goals, motivations, and representative personal characteristics.



What's in a useful persona?

Grounded in reality

Goal-oriented, specific,
and relevant

Tangible and actionable

Generate focus

Interviewing

Define your objectives

Identify and
select participants

Develop interview questions

Conduct the interviews



- General questions
- Needs and challenges
- Product- or service-specific
- Behavioral



Systems thinking

Consider how your customers fit into the larger system in which your project resides

Lean Startup

Emphasizes learning
through build-measure-
learn feedback loops

Clarify user needs
without the risk of
large-scale failures



Borrowing techniques from product management



Market analysis



**Market
segmentation**



**Needs
analysis**



**Sentiment
analysis**



Surveys



Usage statistics



A/B testing

Knowledge check

True or false: You can use Lean Startup techniques to clarify user needs without the risk of large-scale failures



True



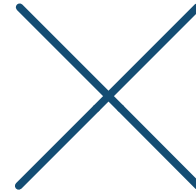
False

Knowledge check

True or false: Personas are developed as an alternative to requirements



True



False



Validate that iteration deliverables meet acceptance criteria

Section 2 of 3

Frequent verification and validation

Agile approaches
were developed to
resolve problems as
early as possible





Test your work

Test it yourself

Get a customer to test it

Does this do what the
customer wants it to do?

**Don't assume
no feedback =
no problems**

More likely it means
no testing or evaluation has
happened



Testing Types

165



**Unit
testing**



**Integration
testing**



**System
testing**



**Acceptance
testing**



**Performance
testing**

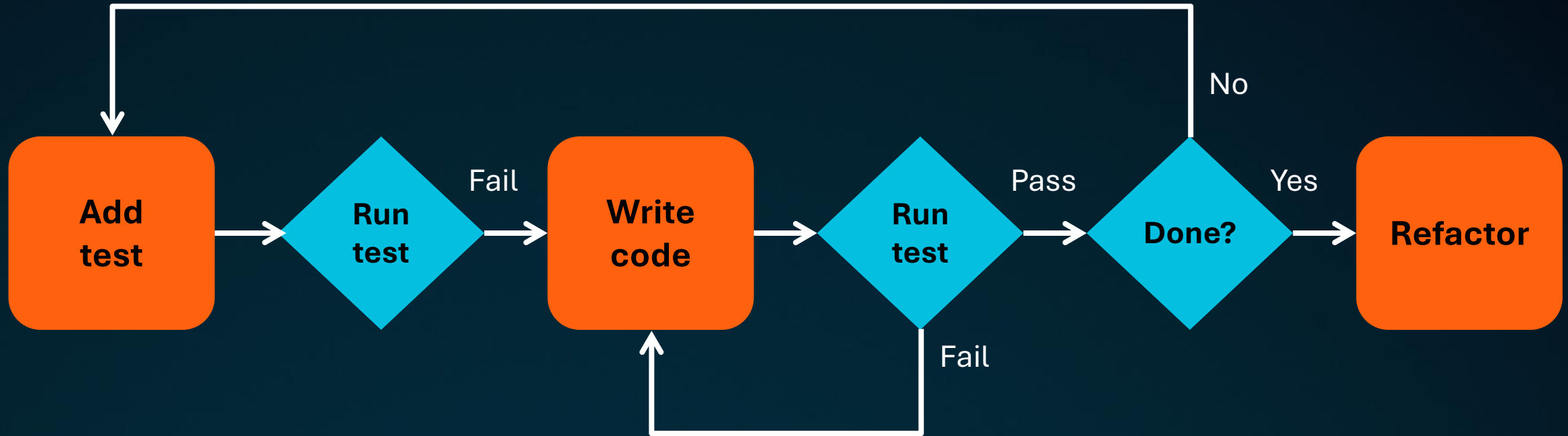


**Security
testing**



**Usability
testing**

Test-driven development (TDD) and test-first development (TFD)



Pros and cons of test-driven development



How will the functionality be used?

Tests in place

Catch defects early

More modular system



Tests written by same developer

Some things are difficult to test

Tests must be maintained

False sense of security

Acceptance test-driven development



Behavior-driven development

Encourages a conversation between the team and the user

Includes a commonly accepted syntax:
given-when-then



Using test-driven development for non-software projects



**Break down
the work**



**Consider how
to test-drive
the work**



**Write the tests
before doing the
work**

Knowledge check

Which of these tests would you perform to see how well your product or service performs under real-world conditions?

A. Usability test

B. Security test

C. Acceptance test

D. Integration test

When you test how well your product or service performs under real-world conditions, you are performing a **usability test**.

Knowledge check

Your team has sent a piece of work for testing. The deadline for testing has passed with no comment from the testers. What can you assume?

A. The piece of work passed the test

B. There are no problems with the piece of work

C. Nothing

D. The piece of work failed the test

No feedback does not mean no problems.



Encourage collaboration between customer and team

Section 3 of 3

Principles of the *Agile Manifesto* (1–4)



1 Our highest priority is to satisfy the customer through early and continuous delivery of valuable software.



2 Welcome changing requirements, even late in development. Agile processes harness change for the customer's competitive advantage.



3 Deliver working software frequently, from a couple of weeks to a couple of months, with a preference to the shorter timescale.



4 Business people and developers must work together daily throughout the project.

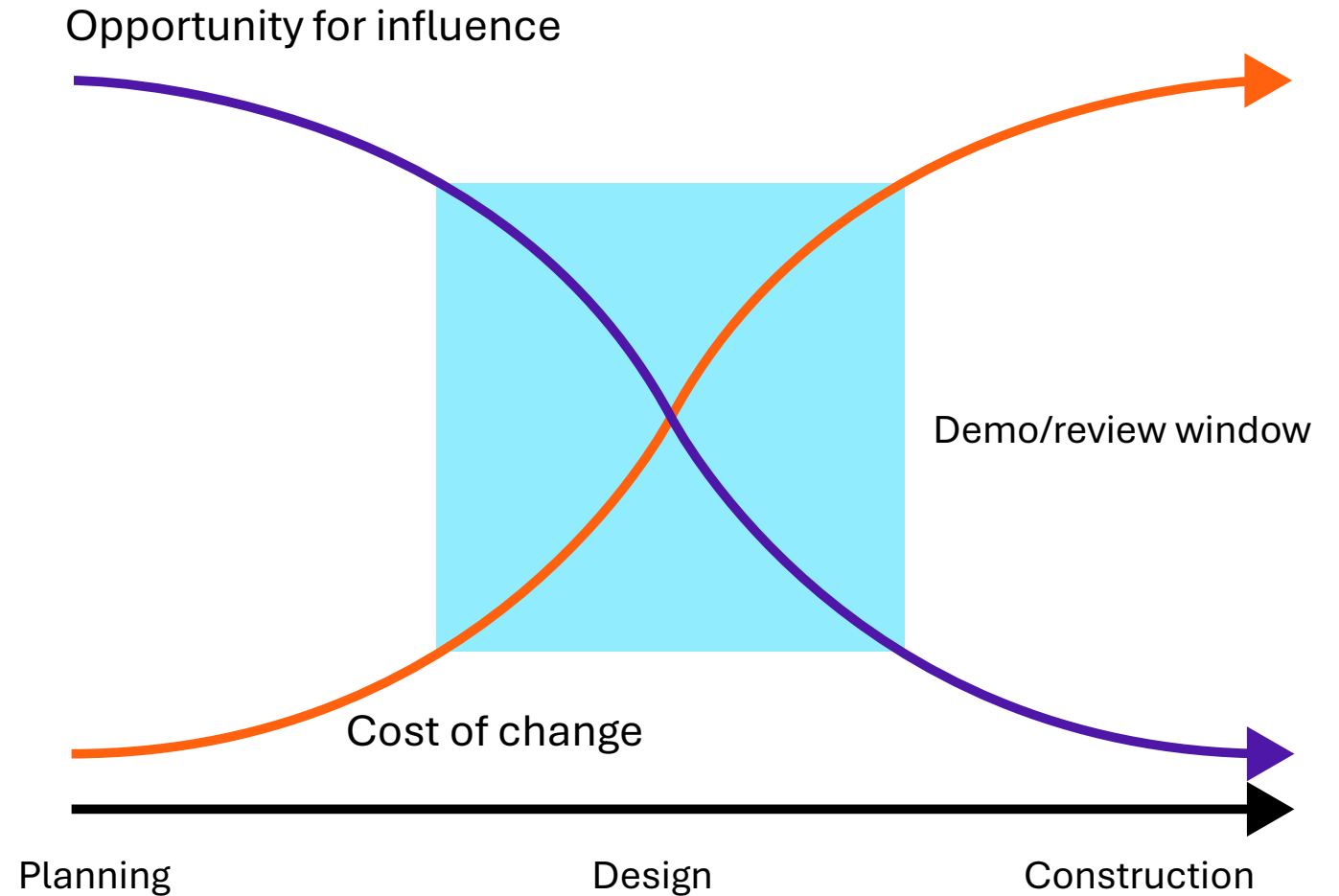
Agile is built on a synergy of

Reviewing at the most efficient point

Initial demos will show incomplete solutions

There will be errors, omissions, and shortcomings

We are reviewing at the most efficient point



Collaboration ensures our team is building what the customer needs

How do you establish a
culture of collaboration?





Regular communication

Frequent meetings

Iterative feedback

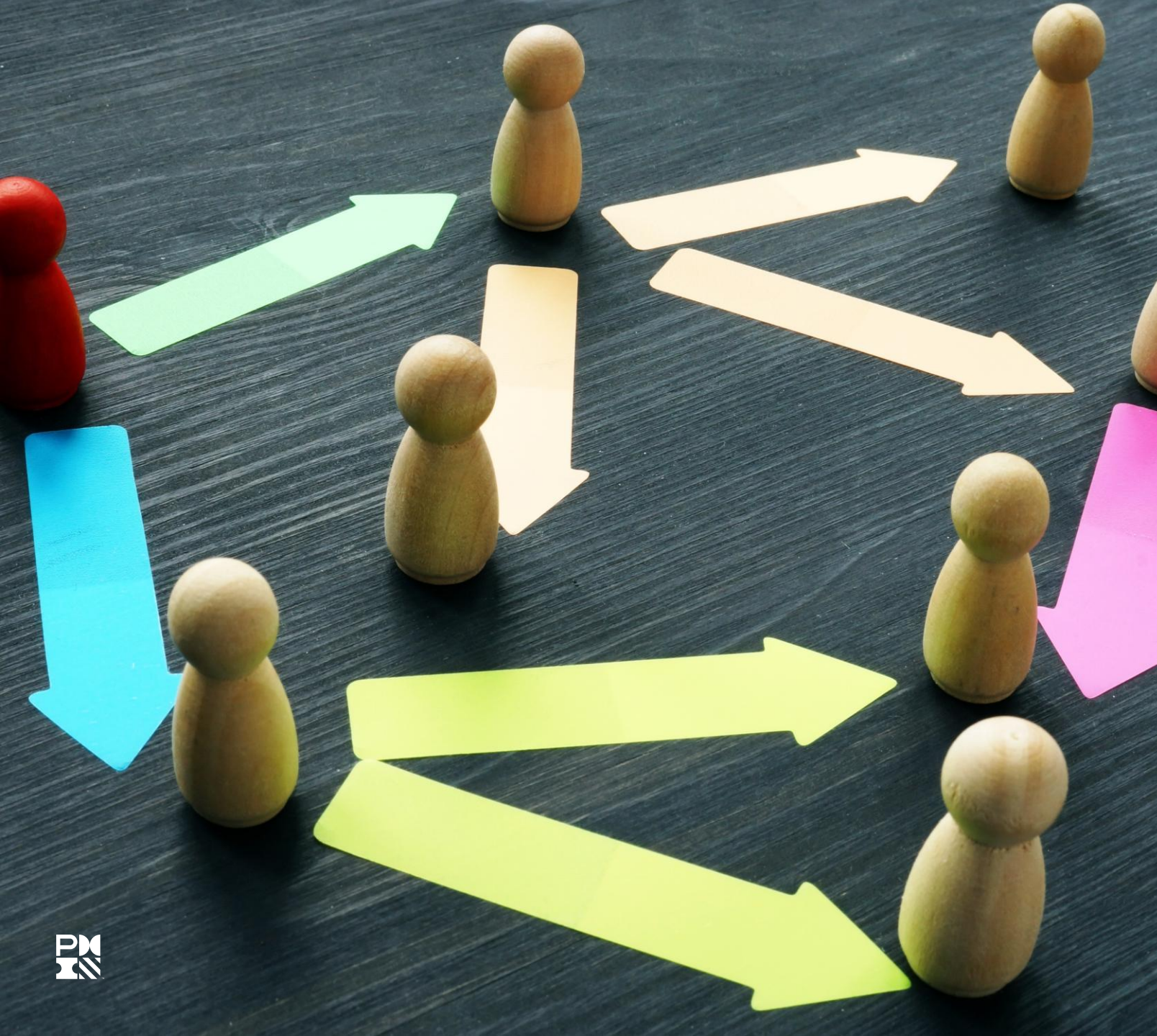
Use collaboration tools

Transparent process

Share progress

Open backlog





Define clear roles and responsibilities

Client representative

Agile roles

Plan collaboratively

Joint planning sessions

User stories

Test criteria





Build trust and rapport

Create a collaborative
environment where the
team and client work
together effectively



**Limit work in
progress at all
levels**



**Shield team
members from
interruptions**



**Use metrics to
analyze and
improve flow**



**Seek early
feedback**



**Manage
agile metrics**



**Manage
impediments
and risk**



**Recognize
and eliminate
waste**



**Perform
continuous
improvements**



**Engage
customers**



**Optimize
flow**



Limit work in progress at all levels

Section 1 of 3

Why should we limit work in progress?

Reducing work in progress reveals the underlying problems in our process



Source:
Goldratt, E. M., & Cox, J. (1992). *The goal: A process of ongoing improvement*. Goldratt Consulting



Other reasons to limit work in progress



**Reduced
multitasking**



**Enhanced
quality**



**Improved
collaboration**



**Faster
feedback loops**



**Better
management**

Limiting work in progress at different levels of the organization

Organization level

Enterprise agility and value stream consulting focus on optimizing value delivery

Department level

Less competition for specialized roles, higher throughput

Team level

Have limits on functions (e.g., testing) often indicated on kanban boards

Individual level

Work on only one or two projects at a time

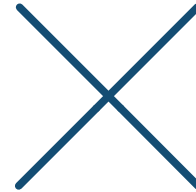
Knowledge check

True or false:

Reducing work in progress can obscure underlying problems in our process



True



False

Knowledge check

True or false:

One advantage of reducing work in progress is that it reduces multitasking



True



False



Shield team members from interruptions

Section 2 of 3

Flow

Flow is the measure of how efficiently work moves through a given process or framework.



**But that is not the Flow we want to talk about.
There is another type of Flow**

What is flow?

The mental state in which a person performing an activity is fully immersed in a feeling of energized focus, full involvement, and enjoyment of the activity



How long does it take to recover from an interruption?

It takes an average of 23 minutes and 15 seconds to get back to the task on which you were working



Source: Mark, G., Gudith, D., & Klocke, U. (2008). The cost of interrupted work. *Proceedings of The 26th Annual CHI Conference on Human Factors in Computing Systems*. <https://doi.org/10.1145/1357054.1357072>

Interruptions increase error rates

Interruptions of as little as
3 seconds doubled the
error rate



Source: Altmann, E. M., Trafton, G., & Hambrick, Z. (2014).
Momentary interruptions can derail the train of thought. *Journal of
Experimental Psychology: General*, 143(1), 215–226.





“What about multitasking ...”

Multitaskers don't pay attention, control their Multitaskers are memory, or switch from one especially vulnerable to job to another as well as irrelevant content those who prefer to complete one task at a time

That's why we want to shield our team from interruptions

We want our team
experiencing that?
as much as possible





Deflection shield

They can act as a deflection shield, protecting the team from a multitude of effective interruptions



Sacrifice one

This is a rotating role

Knowledge check

To shield team members from interruptions, who should respond to requests for external work?

A. Team members

B. Human resources

C. Stakeholders

D. Team lead

The **team lead** acts as a deflection shield, protecting the team from all outside interruptions.

Knowledge check

On agile teams that don't have an effective team lead, a team member steps up to handle requests. How does the team decide who fills this role?

A. It falls to whoever is still standing when the music ends

B. It is decided with a game of rock, scissors, paper

C. It rotates among team members

D. It falls to the most efficient team member

Different members step up to fill the role at different times.



Use metrics to analyze and improve flow

Section 3 of 3

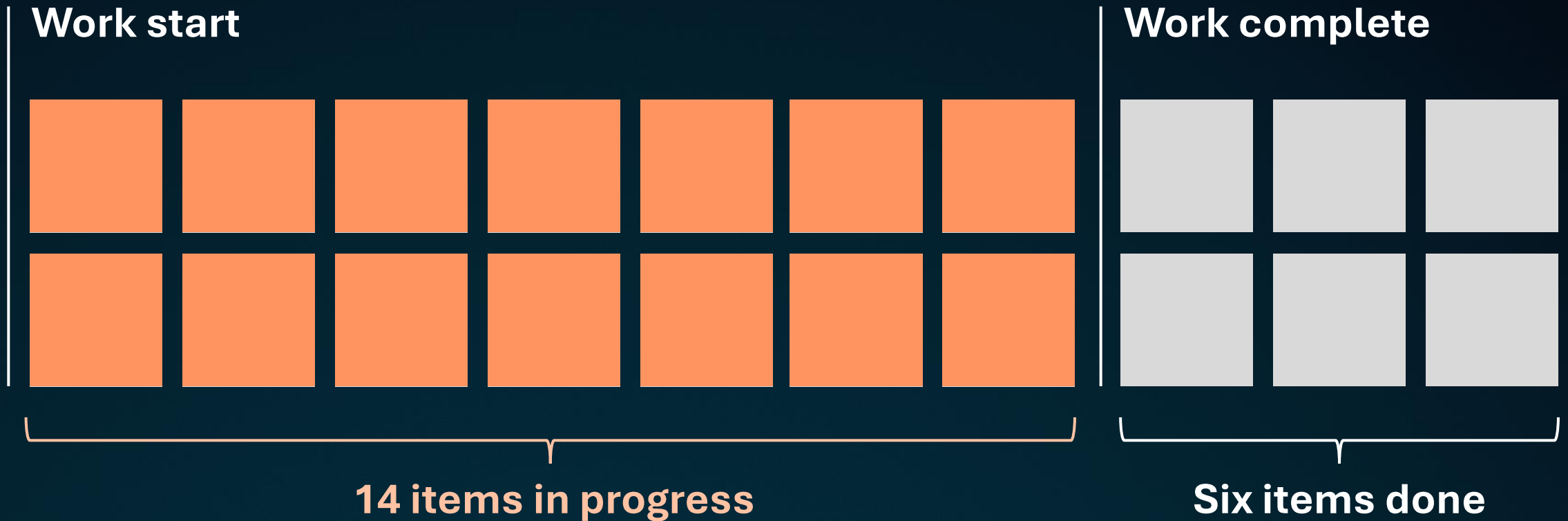
Limiting work in progress

If we're going to limit work
in progress, we should
track it



**Work
in
Progress**

What is work in progress



The amount of work that is currently being processed but not yet completed



Considerations

High work in progress can indicate bottlenecks and inefficiencies

Managing work in progress is crucial for maintaining a smooth workflow

Kanban boards with work in progress limits

This is a strategic decision by the team



Kanban metrics

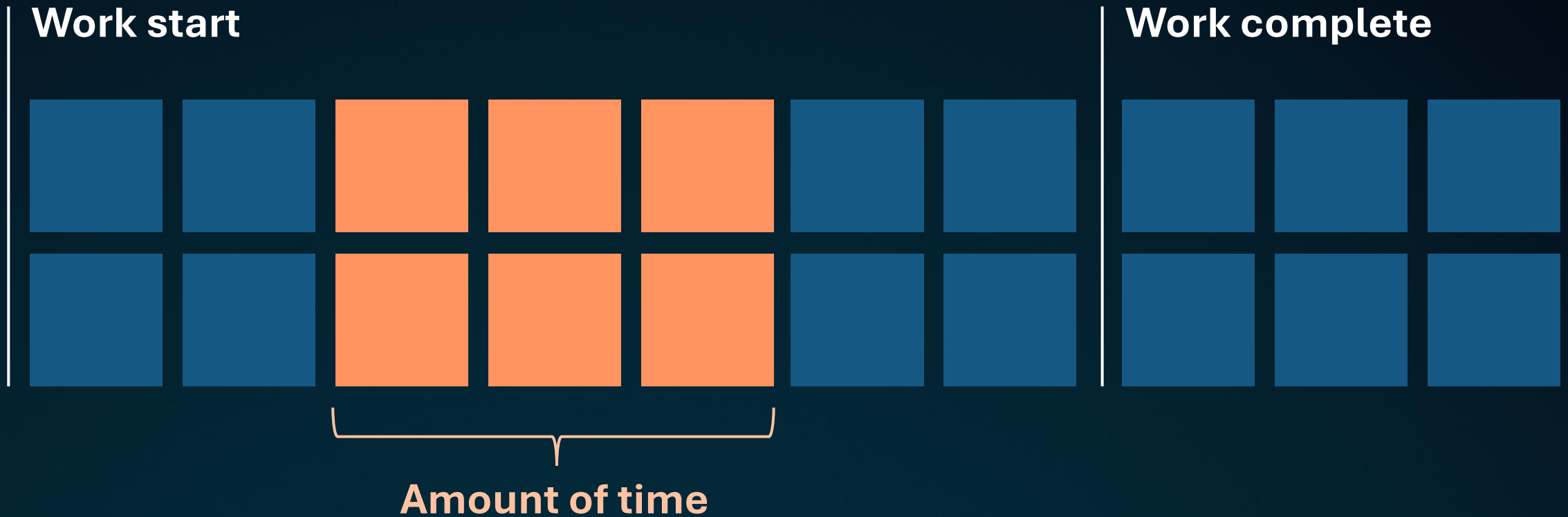


Lead time



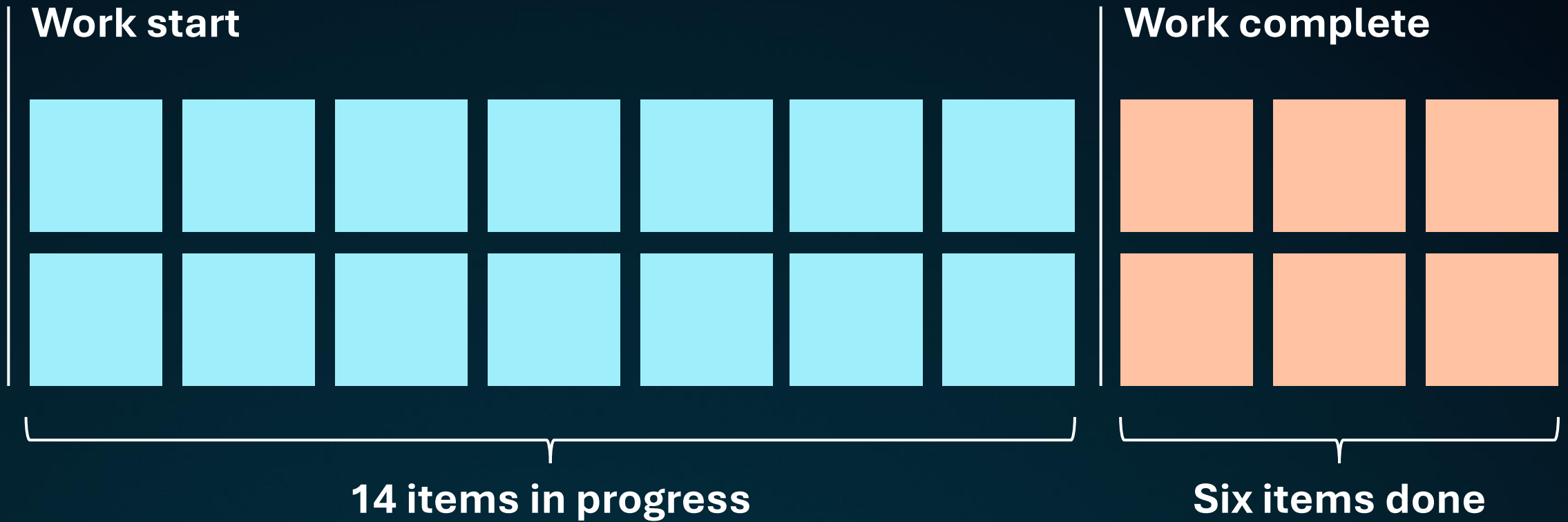
The time between a customer request and the actual delivery

Cycle time



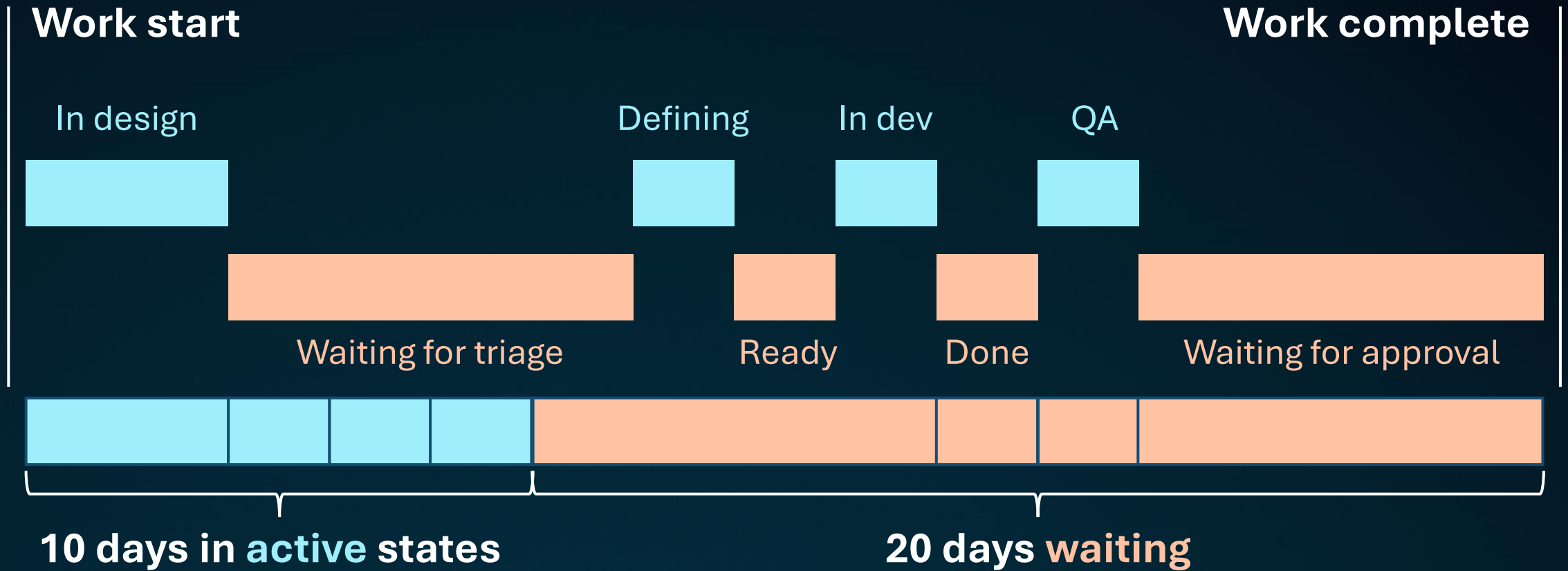
The time it takes to complete a specific task from start to finish

Velocity



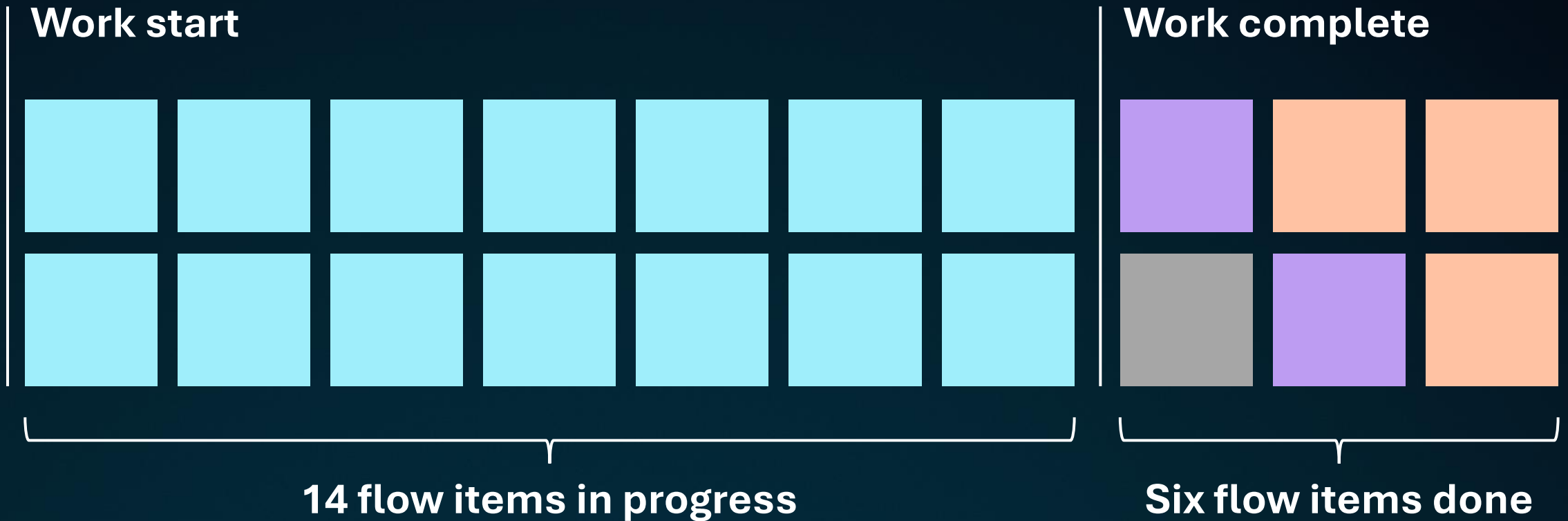
Velocity = six

Process efficiency



$$\text{Process efficiency} = 10 \div 30 = 33\%$$

Flow distribution[®]



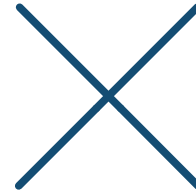
Flow distribution = **50% new features** •
33% defect fixes • **17% task to address debt** • **0% risks**

Knowledge check

True or false: Work in progress is the time it takes to complete a specific task from start to finish



True



False

Knowledge check

True or false: Process efficiency is the ratio of active time versus wait time



True



False

Next Steps

1. Submit Application
2. Create Study Schedule
3. Set Target Date
4. Commit to goal and study plan
5. Schedule/Pay for Exam
6. Test Experience Tips
7. Network/Collaborate



Source: Altmann, E. M., Trafton, G., & Hambrick, Z. (2014).
Momentary interruptions can derail the train of thought. *Journal of
Experimental Psychology: General*, 143(1), 215–226.



CERTIFICATE OF COMPLETION

Percipio Users:

- When attending live,.. always login through Percipio
 - even if you are simultaneously logged in through Zoom.
- This helps ensure Percipio auto generates the certificate at the end.

To check your certificate auto-generated after the last session:

Log into Percipio

- > Select **My Learning** in the upper right corner
- > Select **Learning activity**
- > Select the **"PMI PDU"** tab
- > locate **PMI-ACP ATP Bootcamp**
- > Select the **three vertical dots** to the far right
- > Click **Download Certificate.**

CERTIFICATE OF COMPLETION

PERCIPPIO USERS

- Your **certificate will not auto generate** after the last session ends **IF** the following situations happen:
 1. You didn't log into the Bootcamp through Percipio or the Percipio calendar link for one or more sessions.
 2. You missed too much time in one or more sessions.
 3. Your organization turned off certificates.

Then you will have to manually submit an application for the certificate of completion.

CERTIFICATE OF COMPLETION

NON PERCIPIO USERS and

PERCIPIO USERS (who don't have their certificate auto-generated)

- To manually request your official certificate:
Review the instructions in the Attendance Tracker file, which explains how to **manually request a certificate**. **DO NOT** manually request a certificate until **after the Bootcamp is over** and you have **made up any missed session(s)** if applicable.
- Attendance Tracker Word document file can be found at:
<https://github.com/Skillsoft-Content/PMI-ACP-Replay>
 1. Click <> Code
 2. Download Zip.
 3. Unzip the downloaded .zip file.
 4. Access the files in the folders.

DAILY PMP BOOTCAMP SURVEY



- **Our goal** is to provide the best possible Bootcamp experience for a live streaming webinar, with hundreds of participants.
- For each Bootcamp session,
 - Let us know **what you liked** about the experience – your comments really matter.
 - Please include a thank you **to the mentor(s)** working off camera.
 - If you have **recommendations**, share those too!

We sincerely value your opinion!

LOOK FOR THE SURVEY LINK IN THE CHAT

Survey Scale

- This Scale: 0 not at all likely- 10 extremely likely



On a scale of 0-10, how likely are you to recommend this bootcamp to someone else?

This Scale: 0 not at all likely - 10 extremely likely

0	1	2	3	4	5	6	7	8	9	10
☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐